Maharashtra Engineering Training Academy, Nashik

Professional Exam Division

Practice Question for CEA

Mathematics & Applied Mechanics

सदर सराव प्रश्न हे परीक्षार्थींच्या सोईसाठी व सरावासाठी देण्यात येत आहेत. परीक्षार्थीच्या वारंवार मागणीनुसार सदर प्रश्नांची उत्तरे उपलब्ध करून देण्यात येतआहेत. तरी या सराव प्रश्न व उत्तरे यांची खात्री परीक्षार्थींनी आपल्या स्तरावर तपासून पाहणे. या विषयीच्या कुठल्याही तक्रारीची दखल या कार्यालयामार्फत घेण्यात येणार नाही याची नोंद घ्यावी.

Sr. no.	Question	Correct Option	Option 1	Option 2	Option 3	Option 4
1	Gravitation is inversely related to	4	distance between masses	product of magnitude of masses	direction of masses	square of distance between masses
2	Here f (x) = $\log_2 x + 16^x$ then what is value for f(1/4) ?	3	4	2	0	-1
3	Heron's formula is used for finding _____ of a triangle	2	volume	area	perimeter	none of these
4	How many sides does the triangle have ?	2	1	3	4	no diagonals
5	Hook's law hold good for	1	elasticity	kinetic energy	electricity	none of these
6	If 16^2-16- x^2 = 15, find x	2	16	15	24	30
7	If $3^4 = 9^x$ what is the value of x ?	1	2	3	4	0
8	If an object speeds up from 20m/s to 100m/s in 4s, what is its acceleration?	2	80 m / sec	20 m/ sec ^2	40 m / sec ^2	40 m / sec
9	If $a^x = b^y$, then log b / log a is equal to	1	x/y	y/x	x^y	y^x
10	If body is at rest under action of forces the condition said to be ----	1	static	moving	dynamic	kinematic
11	If log 2 = 0.3010 and log 3 = 0.4771 as base 10 then value of $\log_{10}$ 72 is	1	1.8572	2.0343	1.7782	2.7782
12	If log 900= 2.9542, what is value of log 3 ?	1	0.4771	0.9845	1.4771	0.9542
13	If one angle of a linear pair is obtuse then other angle is	2	obtuse	acute	opposite	none of these
14	If sin x = 3/5 then cos x = ?	2	5/5	4/5	5/4	5/3
15	If the difference between the circumference and radius of a circle is 37 cm, then the area of the circle is - sq. cm	4	111	148	259	154

16	If the lines given by $3x+ky = 2$ and $2x + 6y = 1$ are parallel, then the value of k is	1	9	6	4	3
17	If the mechanical advantage of a simple machine is 3, then the	2	effort is 3 times the output force	output force is 3 times the effort	efficiency is 3%	the work output is 1/3 times the input
18	If the radius of cylinder is halved and height is doubled, then what will be the curved surface area?	2	increase by 1	the same	double	triple
19	If $x = 3t^2 - 2t$, find dx/dt when t = 3 sec	3	10	12	16	0
20	If sum of two number is 4 and difference is 2 , find the numbers ----	2	2,1	3,1	4,6	1,2
21	If y=5x + 1 is line equation then value of slope is equal to----	2	3	5	1/5	1/2
22	In 15 minutes a car whose speed is 40 km/h travels a distance of	1	10 km	20 km	40 km	25 km
23	In a cantilever beam reinforcement is placed ..... beam section	4	@ bottom of	on the both faces	at the ends only	on the top of
24	In a homogeneous body, the whole weight of the body is assumed to be concentrate at point is called	1	centre of gravity of the body	centriod of the body	centre of the body	all of these
25	In all cases, the lines rise from left to right have ------ slopes	2	negative	positive	both of 1 & 2	none of these
26	In gravitational force the value of 'g' in MKS system is …….	1	9.81 m/s^2	981 m/s^2	98.1 m/s^2	none of these
27	In mechanics if input and output forces are equal them MA is equal to ---	2	0	1	-1	0.5
28	In mechanics, full form of MA is	1	Mechanical Advantage	Machine Axle	Major Advantage	Minor Advantage
29	In RCC beam bent up bars are provided @ both ends for	2	bending moment	shear force	self weight	none of these
30	In the sequence 2,5,8,11 ..... find next number	2	12	14	15	17
31	In the short formof UDL, D stands for	2	Deformed	Distributed	Disturbed	Distortion

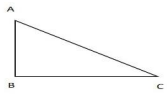
32	In which cases the object does not possess an acceleration or retardation when it moves in	4	upward direction with decreasing speed	downward direction with increasing speed	with constant speed along circular path	with constant speed along horizontal path
33	Instrument used to measure area of irregular shape is …….	3	clinometer	level	planimeter	theodolite
34	Line is made up by ----- number of points	4	two	one	three	infinity
35	Liquid A and liquid B exert same amount of pressure on each other, but density of A is twice density of B. height of liquid B is 10 cm, then height of liquid A would be	1	5 cm	10 cm	20 cm	40 cm
36	Minimum Point required to form a line	1	Two	Three	One	none of these
37	One arg work is equivalent to ……	1	1 dyne-cm	9.81 dyne-cm	1 dyne-meter	1 kg wt-meter
38	Point where whole weight of body acts vertically is called	3	center of mass	mid point	center of gravity	none of above
39	Poisson's ratio is given by …….	1	lateral strain / longitudinal strain	lateral stress / longitudinal strain	stress / strain	none of these
40	Rate of change of angular momentum is equal to	2	force	torque	impulse	linear-momentum
41	Ratio between force of limiting friction and normal reaction is	3	always negative	always zero	constant	variable
42	Semicircle may be called as	3	Quarter sector	half sector	full sector	none of these
43	Shear force is measured in ……..	3	Newton meter	Newton /area	Newton	has no unit
44	Simple pendulum having length l & energy E amplitude A what will be the energy if length of pendulum is reduced to half	3	E/3	E/2	2E	3E
45	Simply supported beam having udl as 4 kg and length 4 then reaction at ends is	1	8 kg	4 kg	16 kg	12 kg
46	Simpson's rule actually gives exact results when approximating integrals of polynomials up to ----- degree.	3	first	second	third	none of these

47	Slope =Rise / Run, then Rise is equal to	1	difference of y coordinates	difference of x coordinates	sum of y coordinates	sum of x coordinates
48	Strain deals with	4	change in shape or size	relative change in shape or size	change in length of object	all of these
49	Strut in trusses is under-----	2	tension	compression	no force	none of these
50	Sum of all angles at a point in a plane is _____	2	180	360	80	0
51	Sum of two sides of a triangle is always ------ than the third side.	3	less	equal	greater	none of these
52	The $(x-4)^2 + (y-3)^2 = 5^2$ representing circle in form of ax^2 + by^2 + cx+ dy +e =0 then value of e is equal to	4	5	3	2	zero
53	The perimeter of square having side length 5 cm is equal to	1	20 cm	35 cm	10 cm	25 cm
54	The unit m/s^2 is used for	4	distance	displacement	velocity	acceleration
55	The velocity is measured in	1	m/sec	m/sec^2	m/sec^3	none of these
56	The angle 90^0 is equal to …….. radians	3	π/3	π/4	π/2	3π/4
57	The area of a triangle is calculated by using the formula ---	2	base times height multiplied by 2	base times height multiplied by 1/2	base times height multiplied by 1/3	base multiplied by height
58	The centre of gravity is the ratio of ---------	1	The product of centroid and weight to the total weight	The addition of centroid and weight to the total weight	The subtraction of centroid and weight to the total weight	The product of centroid and weight to the total mass
59	The diagonals of a rhombus are 8 cm and 10 cm. Then the area of the rhombus is ---	4	64 sq. cm	100 sq. cm	80 sq. cm	40 sq. cm
60	The distance between origin & point P (3,4) is ……	2	7	5	10	12

61	The efficiency of a simple machine is---	1	is always less than 100%	is equal to 100%	is always 50%	is always more than 100%.
62	The equation $x^2 + 6x + 11$ is expressed in the form of $(X+A)^2 + B$ what are the values of A & B	1	3, 2	3, -2	(3, 10)	(-3, -10)
63	The force 30N makes angle 60^0 with X axis then resolution in X axis is	4	$30/\sqrt{2}$	$30\sqrt{2}$	15/2	30/2
64	The force involved in falling apple from a tree is ---	2	contact force	gravitational force	electrostatic force	magnetic force
65	The initial velocity of a body is u. It is under uniform deceleration a. Its velocity v at any time t is given by	3	v = u + at^2	v = u - ½ at^2	v = u - at	v = u.
66	The length and breadth of a rectangle are (4x + 6) m & (3x-4) m respectively. The perimeter of the rectangle is 116 m, Find its area	1	760 m^2	680 m^2	670 m^2	706m^2
67	The line y = mx + c is parallel to y axis then value of m is …..	4	90	0	1	infinity
68	The $\log_{10}30$ is equal to 1.47712 , what is value of log300 with base 10 is	2	0.47712	2..47712	3.47712	none of these
69	The long form of BMD is----	3	Base Moment Diagram	Base Money Diagram	Bending Moment Diagram	Base Moment Distance
70	The pivot point in lever is called as ----	3	chain	pulley	fulcrum	paddle
71	The ratio of the force produced by a machine to the force applied to it and used in the assessing the performance of a machine is called as	2	chemical advantage	mechanical advantage	physical advantage	none of these
72	The ratio of the volumes of two cubes is 729 : 1331. What is the ratio of their total surface areas?	1	81 : 121	09:11	729 : 1331	27 : 121
73	The rectangle having first point as origin & second diagonal point as (3 , 4), then area of rectangle is ----- unit	1	12 sq	7 sq	9 sq	16 sq
74	The shape suspended cable is ……	2	cubical	parabolic	hyperbolic	none of these
75	The sides of a triangle are 5 cm, 12 cm and 13 cm. then its area is ---- sq. meter	3	65	60	0.003	0.3
76	The simple machine that is used to lift heavy objects when they require less force is called is	2	lever	pulley	gear	none of these

77	The slope of speed-time graph gives	3	speed	velocity	acceleration	momentum
78	p	1	1/2	3/2	1/4	3/4
79	The value of equation $(1/2)^{2x+1} = 1$ for x= -1 is equal to	3	1/2	- 1/2	2	-2
80	The value of Log_2 64 is equal to 2 then Log 64 to the base 8 is equal to	4	3	1	5	2
81	The value of sin75 equal to	2	sin(15)	cos(15)	tan(15)	cot(15)
82	The value of 'π' (pi) is given as	2	7/22	22/7	18/7	15/7
83	Three times the square of a natural number is 363, find the number	3	-11	-12	+11	+13
84	To form a triangle , ----- segments are required	3	one	two	three	zero
85	Trapezium is …… type of polygon	4	triangle	hexagon	pentagon	quadrilateral
86	Two lines y = 5x - 5 and y = 2x - 2 meets at point	1	(1,0)	(5,2)	(-3.5)	(0,1)
87	Two lines are perpendicular to each other then angle made is	3	180	270	90	360
88	Two lines make same angle with x alis having different intercepts, then these lines are _____	2	concurrent	parallel	coincident	none of these
89	Two right circular cones of equal curved surface areas have slant heights in the ratio of 3 : 5. Find the ratio of their radii.	3	4 : 1	3 : 5:	5 : 3	04:05
90	Value of cos (35+55) is------	1	0	-1	$1/\sqrt{2}$	1
91	Value of $Cos^2\beta - Sin^2\beta$ is equal to ---	1	Cos2β	Cosβ/2	Sin2β	Sinβ/2
92	What is necessary to get something moving?	2	resistance	force	wedge	screw
93	What is the volume of a hemisphere of radius r?	4	4 π r^3	π r^3	2 π r^3	2/3 π r^3
94	When a force acting on a body displaces it in direction of a force then	3	uniform velocity occurs	uniform speed occurs	work is done	equilibrium take place
95	Which laminas do not have centroid at its geometrical centre?	3	Circle	Equilateral triangle	Right angled triangle	None of these
96	Which is more stable in cube, cylinder & parallelepiped	2	parallelepiped	cube	cylinder	none of these
97	Which is not a scalar quantity ?	3	mass	volume	work	time
98	Which of given formula is correct ?	3	tanA = sinA/cosecA	$cosec^2A + cos^2A = 1$	$cos^2A + sin^2A = 1$	none of these
99	Williot -Mohr-diagram is used to determine deflection in	2	beam only	trusses only	any type structure	rigid frames

100	Write the fraction 7/25 as in percentage	2	25 %	28 %	30 %	10 %
101	Minimum Print required to form a line	1	Two	Three	One	none of these
102	The value of Log_2 32 is equal to	3	6	7	5	2
103	The cantiliver with point load @ free end the maximum B.M will be @	3	Free end	Centre	the fixed end	none of these
104	In mechanics the full form of MA is	3	Mefal Advantage	Mechanical Addition	Mechanical Advantages	Maha Adhar
105	SI unit of volume is	2	m	m^3	m^2	$^1/_m$
106	SI Unit of Force is	1	Newton	Hourse Power	m^2	none of these
107	Value of $Cos^2\beta + Sin^2\beta$ is	1	one	one half	two	two third
108	Sin0 is	1	zero	one	one half	two
109	The intersection of two line is a…..	2	plane	point	prism	triangle
110	y=mx+c, m denotes slope with	1	x axis	y axis	plane	none of these
111	y=mx+c, if C = 0 the line passes throush	1	origin	plane	y axis	none of these
112	Perimeter of circle is given by	3	$2\pi r^2$	πr^2	$2\pi r$	πr^3
113	Surface area of ball is	2	πr^2	$4\pi r^2$	$^4/_3\pi r^3$	$2\pi r^2$
114	Heron's formula is used for finding _____ of a triangle	2	volume	area	perimeter	none of these
115	The long form of BMD is	3	Bending mechanical diagram	Beam moment diagram	Bending moment diagram	none of these
116	For simply supported beem with udl the max BM @ center is	1	$wl^2/_8$	$wl^2/_4$	$wl^2/_2$	$wl^2/_{32}$
117	Sum of all angles at a point in a plane is _____	2	180°	360°	270°	90°
118	A line divides plane in _____ half planes	1	two	three	four	none of these
119	The values of Log_a1 is _____	1	zero	one	two	three

120	Formula of circle in a plane with origin as center & radious r is given by	1	$x^2+Y^2= r^2$	$x^3+Y^3= r^3$	x+y=r	$^x/_2+^y/_2$=r
121	A man standing 30m away from tree & he saw the top of tree then angle made is _____ type.	1	elevation	depression	repose	friction
122	Two lines make same angle with x alis having different intercepts, then these lines are _____	2	cocurent	parallel	coincident	none of these
123	Volume of sphere is given by V= $^4/_3\pi r^3$, the value of π is	1	22/7	7/12	15/7	7/15
124	y=x gives line equation then slope is equal to----	2	two	one	two third	one half
125	The long form of SFD is----	2	shear frame Diagram	shear force Diagram	safe force Diagram	soft from draft
126	 What is sin A	2	AB/BC	BC/AC	BC/AB	AC/AB
127	If tanX = 3/4 then sin X = ?	2	4/3	3/5	5/3	4/5
128	Unit of area is -----	1	m2	N	N/m2	m/s2
129	Unit of acceleration is----	2	N	m/s2	N/m2	m2
130	Modulus of elasticity is given by	1	N/m2	m/N	N/m3	None
131	Energy possessed by body by virtue of its position is called	1	Potential energy	Chemical energy	Mechanical energy	electrical energy
132	The force 30N makes angle 45^0 with X axis then resolution in X axis is	1	30/√2	30√2	15/2	30/2
133	Let point P is given as P(4,4),then angle OP is ---	3	60°	30°	45°	90°
134	sin 2 A = ?	1	2sinA	2 $\sin^2$A	2$\cos^2$A	1/2$\cos^2$A
135	Let f (x) = 16^x, what is f (1/4)	1	2	4	1	8
136	I + tan2 A = ?	3	none of these	$\cot^2$A	$\sec^2$A	san^2A
137	Strut in trussen is under-----	2	tension	compression	No force	none
138	The long form of MI is----------	2	Moment of Idea	Moment of inertia	Minimum Index	Maximum Inertia

139	Line having slope 2 & intercept 5 on Y axis,then line equation is given by-	2	y=5x+2	y=2x+5	y=2/5x+1	y=5/2x
140	A metal rod having diameter d & length l them volume of rod is giving by--	3	$\pi d^2 l$	$\pi/2d^2 l$	$\pi d^2/4l$	πdl
141	simply suported beam having central load w then reaction at ends is	1	w/2	2w	w/4	3w/4
142	$2^4=4^x$, what is the value of X	1	2	1	4	16
143	Every segment has----mid point	2	two	one	three	None
144	Sum of two sides of a triangle Is ------ than the third side.	2	less	more	far	None
145	If the sides of triangle are 3,4,5 what will be the area of triangle	2	20	6	15	10
146	Value of sin (30+60) is------	4	$1/\sqrt{3}$	$\sqrt{3}/2$	$1/\sqrt{2}$	1
147	In triangle,the correct statement is…..	2	it has two abtuse angles	it has two acute angles	All angles more more than 60^0	All angles are less than 60^0
148	Hook's law hold good upto for	1	elasticity	kinetic energy	electricity	none
149	A pilot see down to earth,the angle made is ….. type	2	elevation	depression	acute	abtuse
150	Paralelogram with length l & distance between its two line is h then area is given by	3	lh/2	lh/3	lh	lh/4
151	$\log_a y^m$ =	1	$m\log_a y$	$a\log_m y$	$m\log_y a$	$y\log_a m$
152	$\log_a(x/y)$ =	1	$\log_a x - \log_a y$	$\log_a y - \log_a x$	$y\log_a x$	$x\log_a y$
153	Volume of cone with base rudius r & heigth h is given by ……	2	$1/4\pi r^2 h$	$1/3\pi r^2 h$	$'4/3\pi r^3 h$	$4/3\pi r^2 h$
154	The circle having area A then its perimeter is equal ……	3	$2\sqrt{(A\pi)}$	$\sqrt{(2A\pi)}$	$\sqrt{(A\pi)}$	$\sqrt{(A\pi/2)}$
155	A sphere having volume V & radius r then its surface area is given by ……..	1	$4\pi r^2$	$4/3\pi r^3$	$3\pi r^3$	$3/4\pi r^3$
156	The value of $\sin^2 30$ is …….	3	1/2	3/2	1/4	3/4
157	Two line are perpendicular to each other then angle made is ……	2	180°	90°	270°	360°
158	A line is given by Y = mx + c then slope of perpendicular line is ……	3	2 m	m/2	-1/m	2/m

159	Full form of udl is …….	2	uniformly distorted length	uniformly distributed load	under distance length	uniformly distributed length
160	If $\sin^2A + \cos^2A = 1$, what is value of $\sec^2A - \tan^2A$	1	1	2	1/2	1/4
161	A line make angle Ø with X axis then its slope is …….	2	sinØ	tanØ	cosØ	cotØ
162	SecA = ?	2	1/sinA	1/cosA	1/tanA	1/cota
163	A rope is pulled by a man then rope is under …….	2	compresssion	tension	tortion	none
164	A kite has diagonals d1 & d2 then area is equal to ……..	4	d1d2 / 6	d1d2/3	d1d2/2	d1d2
165	A cube has sides of length l then its volume is ……	2	$l^2/2$	l^3	$l^3/3$	$l^3/4$
166	One radian equal to …….	1	180/π	π/180	π/360	360/π
167	The angle 45^0 is equal to …….. dadians	2	π/3	π/4	π/2	3π/4
168	A line is given of equation 3 x + 5y=15 then slope of line is	2	3/5	-3/5	5/3	-5/3
169	A line given by equation 2 x + 3y= 10, then line intercepts to X axis at a point	1	(5,0)	(2,0)	(3,2)	(2.3)
170	A bar is put under tension & deformed length is l & original length L then strain is given by	2	L/l	l/L	2L/l	2l/L
171	For finding value of sin15 which option is correct	1	sin(45-30)	cos(60-45)	tan(45-30)	cot(45-30)
172	Which is more stable in cube, cylinder & parallelopiped	2	parallelopipe	cube	cylinder	none
173	The parallelopipe has sides 1 mx 2 x 3 m then it will more stable in	3	1m x 3 m	1 m x 2 m	2 m x 3 m	none
174	Simple pendulum having length l & energy E amplitube A what will be the energy if length of pendulum is doubled.	2	E/3	E/2	2E	3E
175	Dimensions of volume given by	2	MLT	$M^0L^3T^0$	ML^2T	MLT^2
176	One jule is equivaleat to …….	1	1 Newton-meter	9.81 Newton-meter	1 dyne-meter	1 kg wt-meter
177	Rate of change of angular momentum is equal to	2	force	torque	impulse	linear-momentur
178	The angles between two forces to make resultant max & minimum respective are	3	0°&90°	180°&90°	0°&180°	90°&270°
179	If the resultant of two forces has same maqnitude as either forces, the angle between the forces is	2	30°	120°	60°	45°
180	The distance travelled by particle is given by $x = 3t^2-2t$ what is magnitude of accerelation @ point of time t = 2 sec	1	10	12	5	3

181	The shape suspended cable @ two ends is ……	2	cubical	elliptical	hyperbolic	none
182	FBD long form is	1	free body diagram	force-body diagram	flip body diagram	none
183	Williot -mohr-diagram is used to defermine deflection in	2	beam only	trusses only	any type structure	rigid frames
184	Joule is unit of	4	power	impulse	momentum	work
185	Coefficient of friction depend on	4	area of contact	nature of surface	strength of surface	all of these
186	Which is not a scalar quantity	3	mass	volume	acceleration	time
187	$(3xy^2)^3$	1	$27x^3y^6$	$9x^2y^4$	$27xy^3$	$9x^3y^2$
188	A polygon which has 6 sides is called as	2	quadrilateral	hexagon	triangle	pentagon
189	Trapesium is …… type of poliygon	4	triangle	hexagon	pentagon	quadrilateral
190	One half of the whole circle is called	3	tragent	sector	semicircle	quadrant
191	The point having x & y values as minus values then it lies in ……. quadrant	4	first	second	fourth	third
192	Area = 1/3 (width of interval) x [(1st + last + 4 sum of even ordinates + 2 sum of odd ordinates)] is called as	3	trapezoidal rule	mid-ordinate rule	simpson's rule	none
193	Instrument used to measure area of irregular shape is …….	3	clinometer	level	planimeter	theodolite
194	In cartesian form point P is denoted by p(X, Y) then its polar form is (r,Q) where Ø is equal to …….	2	$\sin^{-1}y/x$	$\tan^{-1}y/x$	$\cot^{-1}y/x$	$\text{cosec}^{-1}y/x$
195	Area of triagle having sides x, y z is given by √(s(s-x) (s-y) (s-z)) where value of S is……..	4	x+y+z	x+y	x+z	(x+y+z)/2
196	Area c/s of river is calculated by having irregular bank it has different areas @ different height the formula used is ……..	1	Simpson's rule	trapector	mid-ordinate	none
197	Two lines y=3x-5,y=2x-3 meets at point	1	(2,1)	(1,2)	(3.2)	(2,3)
198	The $x^2 + y^2 = r^2$ represents	3	ellipse	prisom	circle	triangle
199	A line intersects circle in …….. points	2	one	two	three	none
200	A boat man saw the top of tree with height h with angle of elevation equal to 30^0 how far he will be away form a tree in terms of height of a tree	1	√3h	√3 / 4 h	4 /√3 h	2/3 h